← Back **Quiz 2: Language Models**
Graded Assignment • 30 min

Due Oct 26, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Higher order N-gram models provide higher performance but suffer from increased sparsity. True or False? 5 / 5 points

Nice work

Corpus

```
<s> some are red </s>

<s> and some are blue </s>

<s> some are old </s>

<s> and some are new </s>
```

2. What is the probability seeing the token "</s>" given the prior 2 tokens "are new", given this corpus. That is, P(</s> | "are", "new"). 5 / 5 points

Assume a trigram model with **no smoothing**. Give a numerical answer not a fraction.

Nice work

3. Given this corpus and a bigram model with no smoothing, what probability would be assigned to the following sentence: 5 / 5 points

```
<s> some are blue </s>
```

Nice work

4. This equation represents an evaluation metric for probabilistic language models. 5 / 5 points

$$H(T) = -\frac{1}{N}\log_2 P_m(T)$$

What's the name of this metric?

Nice work